

UNIT II - Vector Spaces

Notations

$\mathbb{N}$ - Set of all Natural Number

$$= \{1, 2, \dots, \infty\}$$

$$\mathbb{W} = \{0, 1, 2, \dots, \infty\}$$

$$\mathbb{Z} = \{-\infty, \dots, \infty\}$$

$$\mathbb{Z}^+ = \{1, 2, \dots, \infty\} = \mathbb{N}$$

$\mathbb{Z}^*$ - Non negative integers

$$= \{0, 1, 2, \dots, \infty\} = \mathbb{W}$$

$$\mathbb{Q} - \text{Rational Numbers} = \left\{ \frac{p}{q} \mid q \neq 0, p, q \in \mathbb{Z} \right\}$$

$$\mathbb{Q}^* - \text{Irrational Numbers} = \{\sqrt{2}, \pi, \sqrt{5}, \dots\}$$

$\mathbb{R}$ - Real Numbers

$$\{\mathbb{Q} \cup \mathbb{Q}^*\}$$

$$= \{-\infty, \dots, \infty\} \text{ Decimal \& Fractions Included.}$$

$$\mathbb{C} - \{z \in \mathbb{C} / z = x + iy, x, y \in \mathbb{R}\}$$

Groups:

A set G with $*$ is said to be group $(G, *)$ if the following 4 conditions are satisfied.

(i) Closure wrt $*$

$\forall a, b \in A$, then

$a * b = c \in A$ then $*$ is binary operation on A .

A is closed wrt $*$

(ii) Associative wrt $*$

$\forall a, b, c \in A$, If $a * (b * c) = (a * b) * c$

A satisfies Associative property

(iii) Existence of Identity element wrt $*$

$\forall a \in A$, $\exists e \in A$, such that ~~$A *$~~

$$a * e = e * a = a$$

the e is an Identity element of A .

0 is IE wrt $+$.

1 is IE wrt $*$.

(iv) Existence of Inverse Element wrt $*$

$\forall a \in A$, $\exists a^{-1} \in A$ such that $(\exists)$

$$a * a^{-1} = a^{-1} * a = e$$

$(G, *)$ is group

Abelian group.

$$(v) a * b = b * a, \forall a, b \in G$$

Commutative property.

2 MARKS

1. Define Field:-

A Set capital F with $+$ & $\cdot$ is said to be field $(F, +, \cdot)$ if following conditions are satisfied.

(i) $(F, +)$ is an abelian group.

(ii) $(F - \{0\}, \cdot)$ is an abelian group.

(iii) Distributive property of $\cdot$ over $+$

$$(i.e) a \cdot (b + c) = (a \cdot b) + (a \cdot c)$$

$$\forall a, b, c \in F$$

Eg. (i) $(\mathbb{R}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$ are field.

(ii) $(\mathbb{Z}, +, \cdot)$ is field.

2. Define Vector space.

A set V , $+$ scalar multiplication over a field F is said to be vector space if the following conditions must be satisfied. They are,

(i) $(V, +)$ is an abelian group.

$$(v_{s1}, v_{s2}, v_{s3}, v_{s4}, v_{s5})$$

(ii) Closure property of scalar multiplication.
 $\forall v \in V, \forall a \in F \Rightarrow av \in V$

vs7) $\forall v \in V$, then

$$1. v = v \in V$$

vs8) $\forall v \in V, \forall a, b \in \mathbb{F}$,

$$\text{then } a(bv) = (ab)v$$

vs9) Distributive Prty of Scalar Multiplication over Addition.

$$(a+b)v = av + bv$$

$\forall v \in V, a, b \in \mathbb{F}$. then

$$(a+b)v = av + bv$$

vs10) D Prty of SM over Vector Addition.

$\forall u, v \in V, \forall a \in \mathbb{F}$, then

$$a(u+v) = au + av$$

16 MARK :

1. Show that the set of n -tuples of $\mathbb{R}$ real numrs $\mathbb{R}^n$ is a vector space over the field of $\mathbb{R}$ numbers with coordinate wise addition and scalar multiplication as operations.

(00)

Let $V = \{(a_1, a_2, \dots, a_n) / a_i \in \mathbb{R},$

$i = 1, 2, \dots, n\}$. Is V a vector space over $\mathbb{R}$ with operations coordinate wise Add and scalar $\times$

Sol. given, $V = \mathbb{R}^n = \{(a_1, a_2, \dots, a_n) / a_i \in \mathbb{R}, i=1, 2, \dots, n\}$

To check V is an abelian group wrt coordinate wise addition.

Vs1: Closure wrt CW Addition '+'

$$\text{Let } u = (a_1, a_2, \dots, a_n) \in V$$

$$v = (b_1, b_2, \dots, b_n) \in V$$

$$w = (c_1, c_2, \dots, c_n) \in V$$

with $a_i, b_i, c_i \in \mathbb{R}$,

$$i = 1, 2, 3, \dots, n$$

To prove: $u + v \in V$

$$\text{for: } u + v = (a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)$$

$$= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$\in \mathbb{R}^n = V \quad \because a_i + b_i \in \mathbb{R}$$

$\therefore$ Closure property is satisfied.

Vs2: Associative Property:

To prove: $u + (v + w) = (u + v) + w$

$$\text{LHS: } u + (v + w)$$

$$= (a_1, a_2, \dots, a_n) + [(b_1, b_2, \dots, b_n) + (c_1, c_2, \dots, c_n)]$$

$$= (a_1, a_2, \dots, a_n) + [b_1 + c_1, b_2 + c_2, \dots, b_n + c_n]$$

$$= [a_1 + (b_1 + c_1), a_2 + (b_2 + c_2), \dots, a_n + (b_n + c_n)]$$

$$= [(a_1 + b_1) + c_1, (a_2 + b_2) + c_2, \dots, (a_n + b_n) + c_n]$$

$$= (u+v)+w$$

$$= [a_1+b_1, a_2+b_2, \dots, a_n+b_n] + [c_1, c_2, c_3, \dots, c_n]$$

$$= (u+v)+w = \underline{RHS}$$

$\therefore$ Associative Property is satisfied.

V33: Identity Property:

For $u, v \in U \in V$ in vector space, $\exists 0 = (0, 0, \dots, 0) \in \mathbb{R}^n$

$$\Rightarrow u+0 = (a_1, a_2, \dots, a_n) + (0, 0, \dots, 0)$$

$$= (a_1+0, a_2+0, \dots, a_n+0)$$

$$= (a_1, a_2, a_3, \dots, a_n) = u$$

V34: Inverse Property:

For $u \in V$, $\exists u' = (-a_1, -a_2, \dots, -a_n)$

$$\Rightarrow u+u' = (a_1, a_2, \dots, a_n) + (-a_1, -a_2, \dots, -a_n)$$

$$= (a_1-a_1, a_2-a_2, \dots, a_n-a_n)$$

$$= (0, 0, \dots, 0) \in V = \mathbb{R}^n$$

$$= 0 \text{ Inverse}$$

$\therefore$ Identity property satisfied.

V35: Commutative Property

To prove: $u+v = v+u$

$$\underline{LHS:} u+v = (a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)$$

$$= (a_1+b_1, a_2+b_2, \dots, a_n+b_n)$$

$$= (b_1+a_1, b_2+a_2, \dots, b_n+a_n)$$

$$= (b_1, b_2, \dots, b_n) + (a_1, a_2, \dots, a_n)$$

$$= v + u$$

$$= \text{RHS.}$$

$\therefore$ Commutative property is satisfied.

VS6: Closure property of Scalar Multiplication

$$\forall u \in V, \forall d \in \mathbb{R}$$

$$\text{To prove: } du \in V = \mathbb{R}^n$$

$$\text{Let } du = d(a_1, a_2, \dots, a_n)$$

$$= (da_1, da_2, \dots, da_n) \in \mathbb{R}^n = V$$

$\therefore$ It is satisfied.

$$\text{VS7: For } u \in V, 1 \in \mathbb{R} \Rightarrow 1 \cdot u = u$$

$$1 \cdot u = 1 \cdot (a_1, a_2, \dots, a_n)$$

$$= (a_1, a_2, \dots, a_n)$$

$$1 \cdot u = u \in V$$

$\therefore$ It is satisfied.

$$\text{VS8: } \forall \alpha, \beta \in \mathbb{R}, \forall u \in V = \mathbb{R}^n$$

$$\text{To prove: } (\alpha\beta)u = \alpha(\beta u)$$

$$\text{LHS: } (\alpha\beta)u = (\alpha\beta)(a_1, a_2, \dots, a_n)$$

$$= (\alpha\beta a_1, \alpha\beta a_2, \dots, \alpha\beta a_n)$$

$$= [\alpha(\beta a_1), \alpha(\beta a_2), \dots, \alpha(\beta a_n)]$$

$$= \alpha[\beta a_1, \beta a_2, \dots, \beta a_n] = \alpha[\beta(a_1, a_2, \dots, a_n)]$$

$$= \alpha(\beta u) = \text{RHS.}$$

VS9: Distributive property of scalar $\times$ over $+$.

$$\forall \alpha, \beta \in \mathbb{R}, u \in V = \mathbb{R}^n$$

TP: $(\alpha + \beta)u = \alpha u + \beta u$

LHS: $(\alpha + \beta)u = (\alpha + \beta)[a_1, a_2, \dots, a_n]$

$$= [(\alpha + \beta)a_1, (\alpha + \beta)a_2, \dots, (\alpha + \beta)a_n]$$

$$= [\alpha a_1 + \beta a_1, \alpha a_2 + \beta a_2, \dots, \alpha a_n + \beta a_n]$$

$$= [\alpha a_1, \alpha a_2, \dots, \alpha a_n] + [\beta a_1, \beta a_2, \dots, \beta a_n]$$

$$= \alpha(a_1, a_2, \dots, a_n) + \beta(a_1, a_2, \dots, a_n)$$

$$= \alpha u + \beta u$$

$$= \text{RHS} \quad \therefore \text{Satisfied} //$$

VS10: Distributive property of scalar $\times$ over vector $+$

$$\forall \alpha \in \mathbb{R}, u, v \in V = \mathbb{R}^n$$

TP: $\alpha(u + v) = \alpha u + \alpha v$

$\Rightarrow$ LHS: $\alpha(u + v)$

$$= \alpha[(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)]$$

$$= \alpha[(a_1 + b_1), (a_2 + b_2), \dots, (a_n + b_n)]$$

$$= [\alpha(a_1 + b_1), \alpha(a_2 + b_2), \dots, \alpha(a_n + b_n)]$$

$$= [(\alpha a_1 + \alpha b_1), (\alpha a_2 + \alpha b_2), \dots, (\alpha a_n + \alpha b_n)]$$

$$= [\alpha a_1, \alpha a_2, \dots, \alpha a_n] + [\alpha b_1, \alpha b_2, \dots, \alpha b_n]$$

$$= \alpha(a_1, a_2, \dots, a_n) + \alpha(b_1, b_2, \dots, b_n)$$

$$= \alpha u + \alpha v$$

$$= \text{RHS} \quad \therefore \text{It is satisfied.}$$

$\therefore V$ is satisfied all properties.

$\therefore V = \mathbb{R}^n$ It is a vector space.

2) Let $V = \{(a_1, a_2) / a_i \in \mathbb{C}, i=1, 2\}$

if $(a_1, a_2), (b_1, b_2) \in V$ and $C \in \mathbb{R}$

Define $(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 b_2)$

and $C(a_1, a_2) = (Ca_1, a_2)$, V is be a vector space overall with the defined operations.

Sol. given: $V = \{(a_1, a_2) / a_i \in \mathbb{C} = \mathbb{R}, i=1, 2\}$

VS1: Closure property

Let $v_1 = (a_1, a_2) \in V$

$v_2 = (b_1, b_2) \in V$

$a_i, a_2 \in \mathbb{R}$

TP: $v_1 + v_2 \in V$

$v_1 + v_2 = (a_1, a_2) + (b_1, b_2)$

$= (a_1 + b_1, a_2 b_2) \in V$

$\therefore$ CP is satisfied.

VS2: Associative Property:

Let $v_3 = (c_1, c_2) \in V$

TP: $v_1 + (v_2 + v_3)$

$= (v_1 + v_2) + v_3$

AHS: $v_1 + (v_2 + v_3)$

$$= (a_1, a_2) + [(b_1, b_2) + (c_1, c_2)]$$

$$= (a_1, a_2) + [$$